

Atul Ashok Prajapati

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🌐 Thane, India



Education

- 2024 – Present 📖 **Post Doctoral Researcher, University of L'Aquila** in XENONnT experiment and DareDevil Project
- 2019 – 2024 📖 **Ph.D., Gran Sasso Science Institute** Astroparticle Physics
Thesis title: *Multivariate analysis for background rejection in CYGNO/INITIUM experiment.*
- 2018 – 2019 📖 **Junior Research Fellow, Indian Institute of Geomagnetism** in Space Weather
Thesis title: *Development of Artificial Neural Network based 3D Ionospheric Model (ANNIM-3D).*
- 2016 – 2018 📖 **M.Sc. Nuclear Physics, University of Mumbai**
Thesis title: *High Resolution Gamma Spectroscopy using Digital Data Acquisition System.*
- 2013 – 2016 📖 **B.Sc. Physics, University of Mumbai**

Awards

- 2021 📖 **Best Poster Award, 20th International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT-2021)** at Daejeon, South Korea

Research Experience

- 2024 - Present 📖 **PostDoc at University of L'Aquila** L'Aquila, Italy
Peak Bias Reconstruction for XENONnT experiment and Data analysis for DareDevil project
- Developed a correction map for peak reconstruction bias
 - Simulated the data for peak reconstruction bias study using FUSE simulation framework and created a bias map using the simulation, which is used to correct the bias in the science run 2 data.
 - ^{136}Xe $2\nu\beta\beta$ decay signal modeling with MC simulations
 - Using different simulated theoretical models to study $2\nu\beta\beta$ decay spectral shape to probe BSM physics.
 - Data analysis for DareDevil project
 - Analyzed the data from GaAs crystal acquired using different readouts (heat and light channels) and with different radioactive sources.

2019 - 2024

■ **Ph.D. Gran Sasso Science Institute**

L'Aquila, Italy

Multivariate analysis for background rejection in CYGNO/INITIUM experiment.

- Developed Machine learning algorithms (2023)
 - Designed and implemented machine learning-based algorithms aimed at effectively discerning signal from background data, enhancing the precision of data analysis within the CYGNO/INITIUM experiment.
- M-RCNN Model for Track Reconstruction and Particle Analysis (Ongoing)
 - Currently leading the development of an advanced Mask R-CNN (M-RCNN) model for comprehensive track reconstruction, particle identification, and classification, a critical component of the CYGNO/INITIUM project
- Contributed to Gaseous Detector Development (2021-2023)
 - Played a key role in the development and optimization of gaseous detectors, improving their functionality and performance for data collection in the CYGNO/INITIUM experiment.
- Research and Development for INITIUM Experiment (2021-2023)
 - Collaborated on research and conducted precise measurements for the INITIUM project, contributing to the first-ever observation of Negative Ion Drift (NID) at atmospheric pressure using optical readout.

2018-2019

■ **Junior Research Fellow, Indian Institute of Geomagnetism** Navi Mumbai, India
Development of Artificial Neural Network based 3D Ionospheric Model (ANNIM-3D).

- Spearheaded the creation of a sophisticated 3D ionospheric model utilizing neural network methodologies, trained on Ionosonde data and Global Navigation Satellite System (GNSS)-Radio Occultation (RO) data. Demonstrated expertise in accurately predicting electron density in the ionosphere and determining ionospheric height.

2017-2018

■ **Master Thesis, University of Mumbai (UM-DAE-CEBS)**

Mumbai, India

High Resolution γ -ray Spectroscopy using Digital Data Acquisition System.

- Conducted comprehensive characterizations of digital data acquisition systems and analog acquisition systems, ensuring their optimal performance for high-resolution γ -ray spectroscopy.
- Successfully determined the half-life of the metastable state in ^{57}Co through precision measurements using 2 NaI(Tl) detectors, contributing to the project's data analysis.
- Acquired proficiency in constructing decay schemes by measuring ^{133}Ba with the utilization of HPGe and CdTe detectors.

2017

■ **Student Associate Research Program (SARP), Center for Excellence in Basic Sciences (CEBS)**

Mumbai, India

High Resolution γ -ray Spectroscopy using Digital Data Acquisition System.

- Gained hands-on experience with diverse radiation detectors, such as HPGe, Organic Scintillators, and NaI(Tl) scintillators. Proficiently collected and analyzed data from these detectors to support high-resolution γ -ray spectroscopy investigations.

Research Interest

- Deep Neural Networks
- Convolutional Neural Networks (UNets, Res-UNets, Att-UNets, Masked-RCNNs)
- Tree Based Neural Nets
- Dark Matter
- Particle Detectors
- Detector Development
- Space Weather

Skills

Computational Skills

- Coding
 - ROOT, Python, C, C++, MATLAB, $\text{\LaTeX}$
- Image Processing
 - OpenCV, Scikit-image, PIL
- Machine Learning
 - TensorFlow, Scikit-learn, MATLAB, Parallel Programming, Basics of GPU programming for ML
- Operating System
 - MacOs, Linux, Windows, Basic shell scripting

Technical Skills

- Development of Particle Detectors and Proficiency with various Gas Electron Multiplier (GEM) amplification structures
- Image Processing and Particle Track Reconstruction
- Machine Learning Algorithm Development for Particle Identification

Presentations and Posters

Contributed Talks

- Nov-2025
 - Second International Conference on the Physics of the Two Infinities** Tokyo, Japan
 - XENONnT latest results and updates*
 - <https://indico.in2p3.fr/event/35255/contributions/161615/>
- Sep-2023
 - TeV Particle Astrophysics (TeVPA)** Napoli, Italy
 - The CYGNO project for directional Dark Matter searches*
 - <https://agenda.infn.it/event/33457/contributions/207010/>
- June-2023
 - CYGNO Collaboration Meeting** Coimbra, Portugal
 - Low Energy ER/NR discrimination*
 - <https://agenda.infn.it/event/36399/contributions/202612/>
- March-2023
 - ML@INFN: Machine Learning activities at INFN Roma** Rome, Italy
 - Reconstruction and Particle Identification with CYGNO Experiment Using Machine Learning*
 - <https://agenda.infn.it/event/34526/contributions/192115/>
- Feb-2023
 - Science Fair** LNGS, Italy
 - The CYGNO experiment: A directional Dark Matter detector with optical readout*
 - <https://indico.gssi.it/event/469/contributions/1636/>

- Jan-2023  **19th Rencontres du Vietnam (TMEX)** Quy Nhon, Vietnam
The CYGNO experiment: A directional Dark Matter detector with optical readout
<http://vietnam.in2p3.fr/2023/tmex/transparencies/ParallelS1/>
- Dec-2022  **CYGNO Collaboration Meeting** Rome, Italy
Rejecting Electron Background using Machine Learning algorithms
<https://agenda.infn.it/event/33483/contributions/187351/>
- Dec-2021  **20th International Workshop on Advanced Computing and Analysis Techniques in Physics Research** Daejeon, South Korea
Reconstruction and Particle Identification with CYGNO experiment
<https://indico.cern.ch/event/855454/contributions/4596531/>
- Dec-2021  **CYGNO Collaboration Meeting** GSSI, Italy
Electron and Nuclear Recoil Discrimination Studies
<https://indico.gssi.it/event/334/contributions/891/>

Posters

- June-2022  **International conference on Machine Learning for Astrophysics (ML4Astro)** Catania, Italy
Reconstruction and Particle Identification with CYGNO experiment
<https://indico.ict.inaf.it/event/1692/contributions/11272/>
- Dec-2021  **20th International Workshop on Advanced Computing and Analysis Techniques in Physics Research** Daejeon, South Korea
Reconstruction and Particle Identification with CYGNO experiment
<https://indico.cern.ch/event/855454/contributions/4596531/>
- Dec-2021  **62nd DAE-BRNS Symposium on Nuclear Physics** Patiala, India
Performance of Digital data acquisition system with a HPGe detector

Outreach Activities

- 2025  **Science Fair Poster Presentation** LNGS, Italy
Presented a poster for the XENONnT experiment at the LNGS lab during the 2025 Science Fair.
- 2024  **Science Communication at Notte Europea Dei Ricercatori, Sharper 2024** L'Aquila, Italy
Communicated scientific processes to the public during the 2024 "Notte Europea dei Ricercatori: Sharper" event.
- 2023  **Science Fair Presentation** LNGS, Italy
Presented the CYGNO experiment at the LNGS lab during the 2023 Science Fair.
- 2022  **Science Communication at Notte Europea Dei Ricercatori, Sharper 2022** L'Aquila, Italy
Participated in and effectively communicated the science behind various processes to the general public at the "Notte Europea Dei Ricercatori: Sharper" event in 2022.
- 2017/18  **Educational Outreach at Indian Institute of Geomagnetism** Navi Mumbai, India
Presented posters and conducted small physics experiments for school students at the Indian Institute of Geomagnetism in Panvel, Navi Mumbai, in both 2017 and 2018.
- 2015  **Organisational Involvement in GRAVITY Event** Mumbai, India
Contributed to the explanation and organization of physics laboratory experiments during UTKAR-SHA events at SIES College in Sion, Mumbai.

- 2013  **Physics Lab Experiments at UTKARSHA Event** Mumbai, India
Played an active role in organizing the activities of the GRAVITY event conducted by the Physics Department and Physical Society at SIES College in Sion, Mumbai.

Organisation of Conferences/Workshops

- 2025  **Gran Sasso Hands-On 2025 - PhD School On Experimental Astroparticle Physics**
 LNGS, Italy
Tutored and mentored participants in hands-on activities connected to the XENONnT Experiment.
- 2023  **Gran Sasso Hands-On 2023 - PhD Autumn School On Experimental Astroparticle Physics** LNGS, Italy
Served as a tutor for hands-on activities associated with the CYGNO Experiment.
- 2017/18  **Organising Outreach activities at Indian Institute of Geomagnetism** Navi Mumbai, India
- 2017  **Frontiers in light front hadron physics: Theory and experiments (Light cone 2017)**
 University of Mumbai, India
- 2016  **Refresher's course on Mathematical methods in Physics & their Applications** University of Mumbai, India
- 2015  **Organising activities of "Gravity" at UTKARSHA event** SIES, Mumbai, India
- 2013  **Organising activities at UTKARSHA event** SIES, Mumbai, India

Position of Responsibilities

-  **Run Coordinator for XENONnT experiment**
Serving as Run Coordinator for the XENONnT experiment since August 2025.
-  **Student Supervision**
Master's thesis supervision for a project in XENONnT focused on signal and background discrimination using ML.
-  **Shifter for XENONnT Experiment (In-Person and Remote)**
Contributed to XENONnT data-taking and calibration activities on-site and remotely at LNGS.
-  **Replacing MagPump for XENONnT**
Participated in replacing and setting up of the magnetic pump for XENONnT.
-  **LIME Detector Data Shifter (In-Person and Remote)**
Demonstrated accountability by actively participating in data-taking operations with the LIME detector while stationed both in-person and remotely at the underground facility of LNGS.

Schools , Conferences and Workshops Attended

- June-2025  **XENONnT Collaboration Meeting** University of L'Aquila, Italy
- June-2025  **XENONnT analysis workshop** LNGS, Italy
- Dec-2024  **XENONnT Collaboration Meeting** LNGS, Italy
- Dec-2024  **XENONnT analysis workshop** LNGS, Italy
- 2022  **INFN School of Statistics** Paestum, Italy
- 2022  **SiPM Radiation: Quantifying Light for Nuclear, Space and Medical Instruments under Harsh Radiation Conditions** Remote

2021	📖	Quantum Sensors for Fundamental Physics, QSFP	<i>Remote</i>
2021	📖	4th International School on Deep Learning, DeepLearn	Gran Canaria, Spain
2021	📖	School on Underground Physics, SOUP	<i>Remote</i>
2019	📖	Inspiring the Minds of Post Graduates for Research in Earth and Space Sciences (IMPRESS-2019)	EGRL, India
2017	📖	Frontiers in Light Front Hadron Physics: Theory and Experiments (LIGHT CONE) Mumbai, India	
2016	📖	Refresher's course on "Mathematical Methods in Physics and their Applications" Mumbai, India	
2016	📖	Summer School in Physics -SSP2016	Pune, India
2014	📖	Ideas and Application in Physics workshop	SIES, Mumbai, India
2013	📖	Workshop on Nano-particles and MATLAB applications	SIES, Mumbai, India